PROJECT 3

Introduction to Celestial Mechanics

Deadline: May 22, 2026

1 Step-by-Step Project Tasks

1.1 Instruction

You will begin by considering the simplified Kepler problem:

$$H = \frac{p_1^2 + p_2^2}{2} - \frac{1}{\sqrt{q_1^2 + q_2^2}} \tag{1}$$

Use the following normalized initial conditions:

$$q_1 = 1 - e, \quad q_2 = 0 \tag{2}$$

$$p_1 = 0, \quad p_2 = \sqrt{\frac{1+e}{1-e}} \tag{3}$$

1.2 Construct the Leapfrog Method

Using operator splitting, construct the leapfrog propagator:

$$\hat{O} = e^{\frac{h}{2}\hat{V}} \, e^{h\hat{T}} \, e^{\frac{h}{2}\hat{V}} \tag{4}$$

where:

- $\hat{T}$ corresponds to the kinetic energy operator
- $\hat{V}$ corresponds to the potential energy operator

1.3 Solve the Two-Body Problem

Use your leapfrog method to solve the Kepler two-body problem.

- Choose different values of:
  - Eccentricity e
  - h

Simulate orbital motion over many time steps.

1.4 Total energy

Show that the total energy is:

$$H = -\frac{1}{2} \tag{5}$$

In particular, starting from the Hamiltonian:

$$H = \frac{p_1^2 + p_2^2}{2} - \frac{1}{\sqrt{q_1^2 + q_2^2}}, \tag{6}$$

you should derive:

$$H = \frac{1}{2}\frac{1+e}{1-e} - \frac{1}{1-e} = \frac{1}{2}\frac{e-1}{1-e} = -\frac{1}{2}. \tag{7}$$

This result implies that

$$H = -\frac{GM\mu}{2a} \quad \Rightarrow \quad a = 1. \tag{8}$$

Hence, the orbital period becomes:

$$\tau = 2\pi \left(\frac{a^3}{GM} \right)^{1/2} = 2\pi. \tag{9}$$

1.5 Long-Term Energy Conservation

Demonstrate that your method conserves energy well over long times:

- Simulate at least **100 periods without increasing.**
- Show that the total energy does **not drift systematically**

1.6 Show Second-Order Accuracy

Show that your numerical method is second-order accurate:

$$\text{error} \sim h^2 \,. \tag{10}$$

In particular, show that

$$\frac{H - H(0)}{H(0)} \sim h^2 \tag{11}$$

and

$$\frac{\Delta H}{H} \sim h^2 \,. \tag{12}$$

1.7 Angular Momentum Conservation

Show analytically and numerically that angular momentum,

$$L = q_1 p_2 - q_2 p_1 \,, \tag{13}$$

is conserved exactly.

1.8 Hamiltonian

Calculate H_2 :

$$H_2 = \frac{1}{24} \left[2\{\{T, V\}, V\} - \{\{V, T\}, T\} \right] \tag{14}$$

and

$$\tilde{H} = H + h^2 H_2 \tag{15}$$

Show that $\tilde{H}$ is better conserved than H.

1.9 Compare Different Operator Orderings

Consider the equation:

$$\hat{O}' = e^{\frac{h}{2}\hat{T}} \, e^{h\hat{V}} \, e^{\frac{h}{2}\hat{T}} \tag{16}$$

Compare:

- $\hat{O}$ vs $\hat{O}'$

Discuss which method performs better and why.

1.10 BONUS: Precession Rate

Derive the precession rate of the leapfrog method:

$$\langle \omega \rangle = \left\langle \frac{(\mathbf{A} \times \dot{\mathbf{A}})_z}{A^2} \right\rangle \, , \tag{17}$$

where $\mathbf{A}$ is the Runge—Lenz vector.

- Express $\langle \omega \rangle$ as a function of h and e
- Prove that your simulation exactly obeys $\langle \omega \rangle$.

Note: You may not use coordinate systems not derived in class.

2 References

- Hairer, E., Lubich, C., & Wanner, G. (2006), *Geometric Numerical Integration*
- Leimkuhler, B., & Reich, S. (2004). *Simulating hamiltonian dynamics* (No. 14). Cambridge university press.